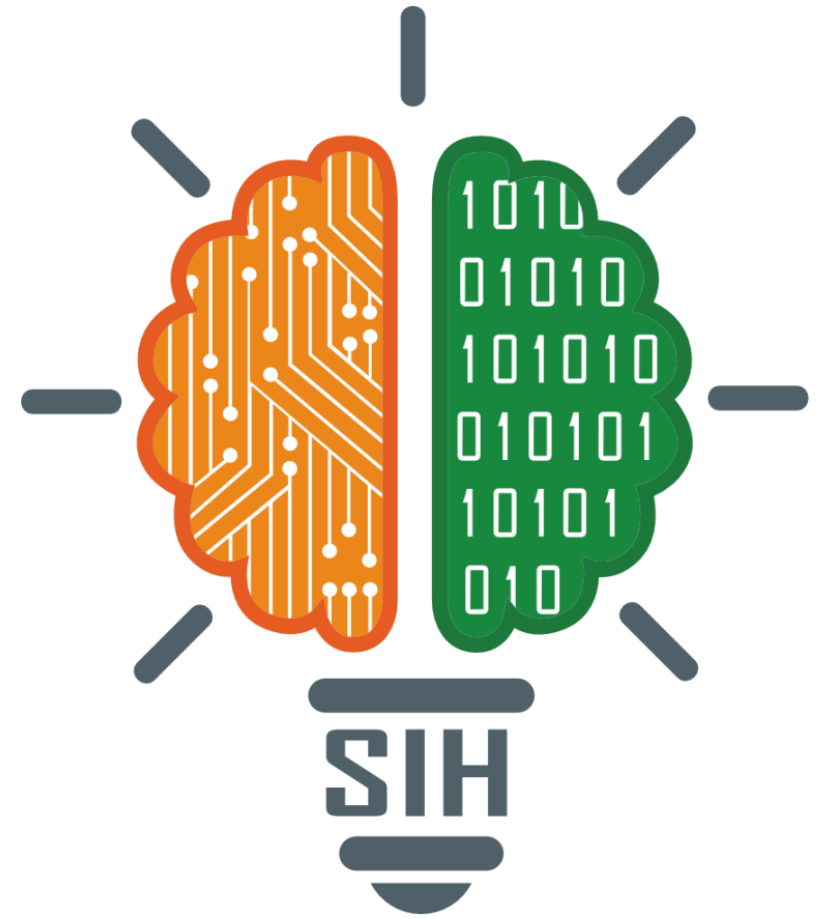


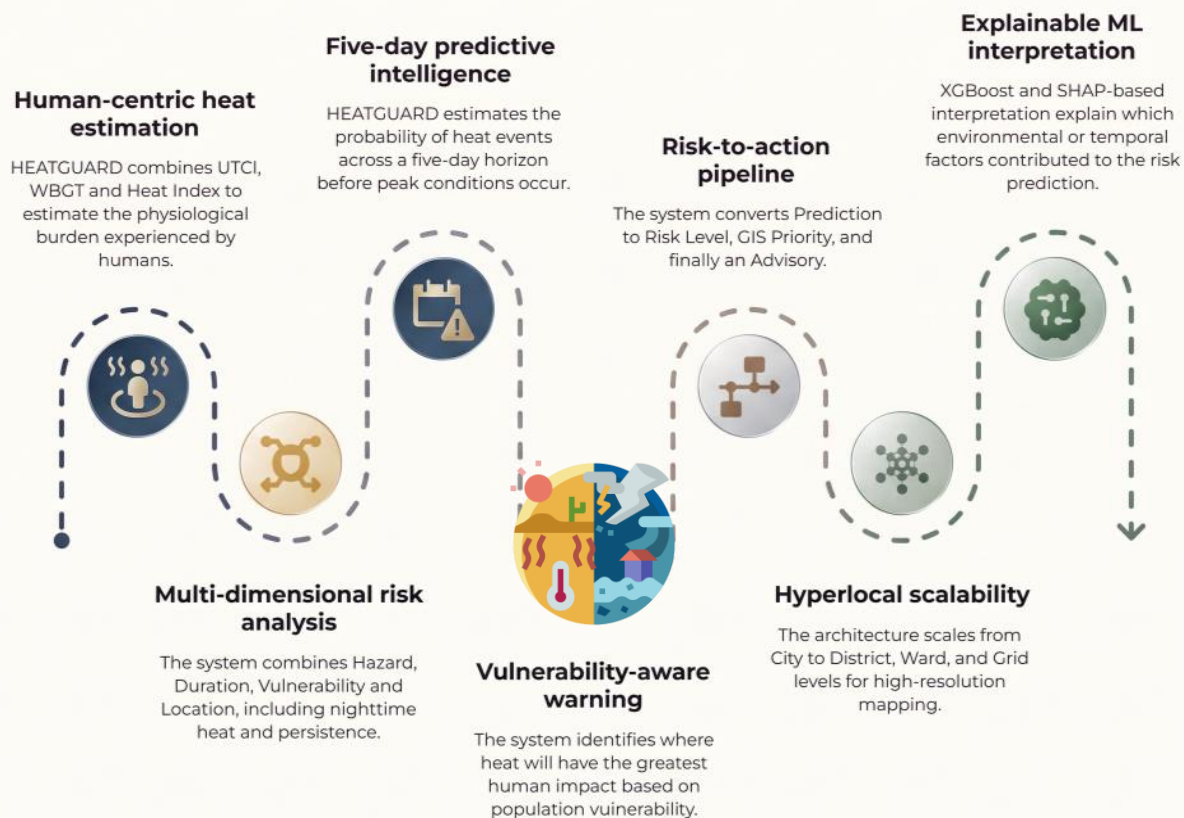
TITLE PAGE

- **Problem Statement ID – 26083**
- **Problem Statement Title – Extreme Heatwave
Early Warning and Human Thermal Stress
Index**
- **Theme – Disaster Management**
- **PS Category – Software**
- **Team Name – Code911**



HEATGUARD: From predicting heat to predicting human impact

HEATGUARD: From predicting heat to predicting human impact



Human-centric estimation

Combines UTCI, WBGT and Heat Index to estimate physiological heat burden.

Five-day prediction

Estimates heat-event probability across a 1–5 day horizon, before peak conditions occur.

Multi-dimensional risk

Combines hazard, duration, vulnerability and location, including nighttime heat and persistence.

Vulnerability-aware

Identifies where heat may have the greatest human impact based on population vulnerability.

Risk-to-action pipeline

Converts prediction into Risk Level, GIS Priority and finally an advisory.

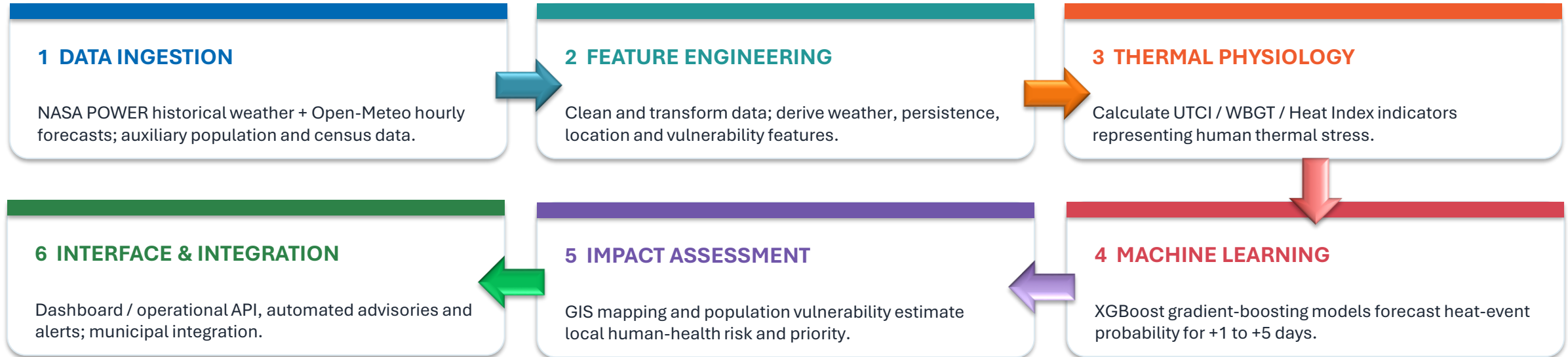
Explainable & scalable

XGBoost + SHAP explain contributing factors; GIS mapping scales city → district → ward → grid.

Outcome: localized, actionable heat-health warnings for safer communities.

Made with  Napkin

HEATGUARD system workflow



TECH STACK



Continuous data update & model retraining

Tech stack: Python • pandas • NumPy • XGBoost • scikit-learn • GeoPandas • Folium • Leaflet • matplotlib • Plotly • Streamlit • FastAPI • Jupyter • Docker

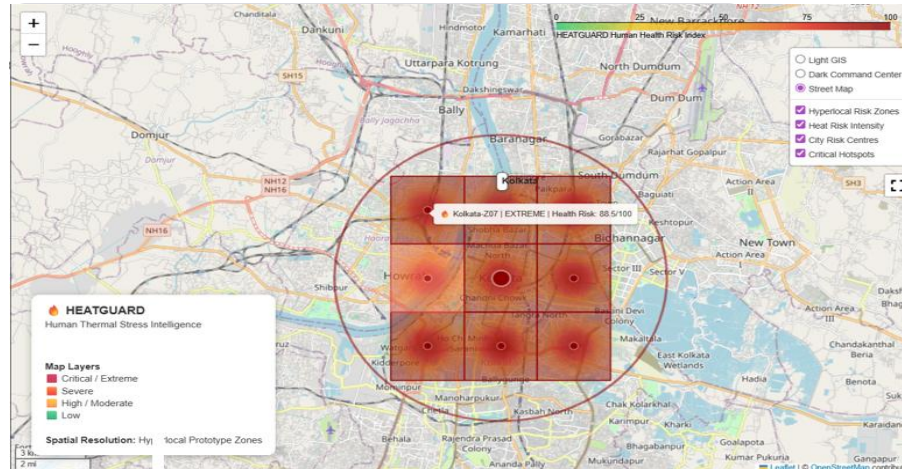
Live GIS Early-Warning Dashboard

Every city's peak 5-day risk, plotted on an interactive India map — the working prototype

● LOW
 ● MODERATE
 ● HIGH
 ● SEVERE
 ● EXTREME

City	Peak Day	Risk Probability	Warning
Delhi	+1 day	99.76%	EXTREME
Kolkata	+4 day	98.76%	EXTREME
Ahmedabad	+3 day	98.64%	EXTREME
Mumbai	+3 day	96.80%	EXTREME
Srinagar	+5 day	0.38%	LOW

Peak 5-day forecast, latest available run across all five cities — direct notebook output.



DASHBOARD FEATURES

- Colour-coded city markers by warning tier
- Click-through popup: peak day, heat probability, vulnerability score and human-health risk
- Tooltip summary for rapid scanning
- Exportable, shareable HTML; deployable on any device without installation

Technical & operational

Advanced ML/DL algorithms; real-time API integration; user-friendly interfaces; scalability across regions.

Data accuracy issues

Incomplete or inaccurate data can cause discrepancies; requires real-time data integration.

Market & sustainability

Demand for precision risk information; supports sustainable, climate-resilient planning.

Data reliability concerns

Region-specific discrepancies require validation and ongoing data-quality checks.

From heat prediction to human protection



AI-POWERED HEATWAVE RISK ASSESSMENT & EARLY WARNING SYSTEM

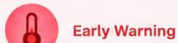
From Data to Action. From Heat Prediction to Human Protection.

A Cooler,
Healthier, More
Resilient India

People
Places
Preparedness
Safer Communities

IMPACT AND BENEFITS

From Heat Prediction to Human Protection



Early Warning

- Predicts heat-event probability 1–5 days ahead
- Detects sustained heat and dangerous thermal stress



Health-Centric Risk

- Goes beyond temperature using UTCI, WBGT & Heat Index
- Combines environmental risk with population vulnerability



Hyperlocal Decision Support

- GIS-based zone/ward-ready risk mapping
- Identifies high-risk hotspots for targeted intervention



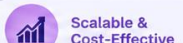
Actionable Response

- Converts risk levels into automated heat-health advisories
- Supports cooling centres, hydration campaigns and outdoor-worker protection



Better Preparedness

- Helps health facilities anticipate periods of increased heat stress
- Enables authorities to prioritize elderly, children and outdoor-workers



Scalable & Cost-Effective

- Uses publicly accessible weather data and open-source technologies
- Can scale from city → ward → district → state

THE CHALLENGE

- Rising temperatures & frequent heatwaves
- Vulnerable populations at higher risk
- Strain on health systems and urban services



THE OUTCOME

- Proactive, localized heat-health preparedness
- Reduced health risks and saved lives
- More resilient and climate-ready communities



KEY IMPACT

HEATGUARD enables authorities to move from reactive heatwave response to **proactive, localized** and **human-centric** heat-health preparedness.

People Protected

Data-Driven Decisions

Stronger Communities

A Cooler, Healthier India

Beat the Heat. Build a Safer Tomorrow.

Early warning

Predicts heat-event probability 1–5 days ahead; detects sustained heat and dangerous thermal stress.

Health-centric risk

Combines UTCI, WBGT and Heat Index with environmental risk and population vulnerability.

Hyperlocal decision support

GIS-based zone/ward-ready mapping identifies high-risk hotspots for targeted intervention.

THE CHALLENGE

Rising temperatures and frequent heatwaves • vulnerable populations at higher risk • strain on health systems and urban services



ACTIONABLE RESPONSE

Automated heat-health advisories; cooling-centre and hydration campaigns; protection for outdoor workers.



THE OUTCOME

Proactive localized preparedness • reduced health risks and saved lives • more resilient, climate-ready communities.

Additional benefits: better preparedness for health facilities • prioritization of elderly people, children and outdoor workers • scalable city → ward → district → state • public weather data and open-source technologies

KEY SCIENTIFIC & TECHNICAL REFERENCES

1. NASA POWER — Historical Weather Data

Historical hourly temperature, humidity, wind, solar radiation and other meteorological variables used for model development.

[NASA POWER API Documentation](#)

2. Open-Meteo — Weather Forecast API

Operational hourly weather forecasts feeding the 1–5 day HEATGUARD prediction pipeline.

[Open-Meteo API Documentation](#)

3. UTCI — Universal Thermal Climate Index

Scientific basis for human thermophysiological response using temperature, humidity/water vapour, wind and mean radiant temperature.

[UTCI Official Research Portal](#)

4. ISO 7243:2017 — WBGT

International standard for heat-stress assessment using the Wet Bulb Globe Temperature index.

[ISO 7243:2017](#)

5. XGBoost — Machine Learning Forecasting

Gradient-boosting framework for five +1 to +5 day heat-event prediction models.

[XGBoost Documentation](#)

6. SHAP — Model Explainability

Interprets which engineered weather and thermal features contribute to model predictions.

[SHAP Documentation](#)

Data → Insights → Action → Safer Communities